

Number Theory and Cryptography

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Number Theory and Cryptography

Number theory is the study of numbers.

The numbers are mostly integers (positive).

Divisibility theory: Positive integers can be partitioned into three basic classes.

- Unit
- Primes
- Composites

Divisibility

Let a and b be integers with $a \neq 0$, then a divides b if there exists an integer k such that $b = k \times a$

- When a divides b then we say that
 - a is a factor/divisor of b
 - b is a multiple of a

Notation: $a|b$

Example: $3|9$, $25|100$

- $a|b$, then $\frac{b}{a}$ an integer.
- a does not divide b then we write $a \nmid b$

Properties

Let a , b , and c be integers with $a \neq 0$, then

- If $a|b$ and $a|c$, then $a|(b + c)$
- If $a|b$, then $a|bc$
- If $a|b$ and $b|c$, then $a|c$
- If $a|b$ and $a|c$, then $a|(mb + nc)$ for integers m and n .

Division theorem

When an integer is divided by a positive integer, then there is **remainder** and **quotient**.

Division theorem: Let a and d be two integers with $d > 0$ then there exist integers q and r such that

$$a = qd + r \text{ where } 0 \leq r < d$$

Example: $19 = 3 \cdot 6 + 1$

- $d \Leftarrow$ divisor
- $a \Leftarrow$ dividend
- $q \Leftarrow$ quotient
- $r \Leftarrow$ remainder

We define **div** and **mod** as $q \equiv a \text{ div } d$ $r \equiv a \text{ mod } d$

Congruence relation

Let a and b be integers and m be a positive integer, then a is congruent to b modulo m if $m|(a - b)$.

- Notation: $a \equiv b \pmod{m}$
- m is called modulus
- $a \equiv b \pmod{m}$ is called congruence
- When a is not congruent to b modulo m then we write.
 $a \not\equiv b \pmod{m}$

Congruence properties

Theorem: Let m be a positive integer and a and b be integers. a is b congruent modulo m iff there is an integer k such that $a = b + km$.

Proof: (if part:)

- $a = b + km$
- $a - b = km$
- $m | (a - b)$
- $a \equiv b \pmod{m}$

(only if part:)

- $a \equiv b \pmod{m}$
- $m | (a - b)$
- $a - b = km$
- $a = b + km$

Congruence properties

Theorem: Let m be a positive integer and a, b, c, d be integers. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then

- $(a + c) \equiv (b + d) \pmod{m}$
- $ac \equiv bd \pmod{m}$

Proof:

- $a \equiv b \pmod{m}$
- $m \mid (a - b)$
- $a - b = km$
- $a = b + km$
- Similarly, $c = d + \ell m$
- $a + c = (b + d) + (k + \ell)m$
- $a + c = (b + d) + tm$

Other part is similar

Congruence properties

Theorem: Let m be a positive integer and a, b be integers. If $a \equiv b \pmod{m}$ then

- $(c \cdot a) \equiv c \cdot b \pmod{m}$, where c is any integer.
- $(c + a) \equiv c + b \pmod{m}$, where c is any integer.

Proof: H.W.

Question: Why is the division not true?

H.W.

Congruence properties

Theorem: Let m be a positive integer and a, b be integers. Then

- $(a + b) \bmod m \equiv ((a \bmod m) + (b \bmod m)) \bmod m$
- $(ab) \bmod m \equiv ((a \bmod m)(b \bmod m)) \bmod m$

Proof: H.W.

Integer representation

Decimal: Base 10 representation. (Digits: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$)

$$(2105)_{10} = 2 \cdot 10^3 + 1 \cdot 10^2 + 0 \cdot 10^1 + 5 \cdot 10^0$$

$$(87)_{10} = 8 \cdot 10^1 + 7 \cdot 10^0$$

Binary: Base 2 representation. (Digits: $\{0, 1\}$)

$$(87)_{10} = (1010111)_2$$

$$(1010111)_2 = 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$

Octal: Base 8 representation. (Digits: $\{0, 1, 2, 3, 4, 5, 6, 7\}$)

$$(87)_{10} = (127)_8$$

$$(127)_8 = 1 \cdot 8^2 + 2 \cdot 8^1 + 7 \cdot 8^0$$

Hexadecimal: Base 16 representation. (Digits: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F\}$)

$$(87)_{10} = (57)_{16}$$

$$(57)_{16} = 5 \cdot 16^1 + 7 \cdot 16^0$$

Base b representation

Take any positive integer $b > 1$ as a base to represent any number.

b and n be integers where $b > 1$ and $n > 0$.

n can be expressed uniquely as,

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b^1 + a_0 b^0$$

where k is an integer and $k \geq 0$, and $a_1, a_2, \dots, a_k$ are non-negative integers that are $< b$ and $a_k \neq 0$.

$a_1, a_2, \dots, a_k$ are called base- b digits.

n is denoted as,

$$(a_k a_{k-1} \dots a_1 a_0)_b$$

Construction of base b representation

Algorithm 1 Representation(n, b)

$tmp = n;$

$k = 0$

while $tmp \neq 0$ **do**

$a_k = tmp \bmod b$

$tmp = tmp \operatorname{div} b$

$k = k + 1$

return ($a_{k-1}a_{k-2} \dots a_1a_0$)

Example: $(17)_{10}$

$$a_0 \equiv 17 \bmod 2 = 1$$

$$tmp = 8$$

$$a_1 \equiv 8 \bmod 2 = 0$$

$$tmp = 4$$

$$a_2 \equiv 4 \bmod 2 = 0$$

$$tmp = 2$$

$$a_3 \equiv 2 \bmod 2 = 0$$

$$tmp = 1$$

$$a_4 \equiv 1 \bmod 2 = 1$$

Number of digits in a representation

Number of digits in a representation:

Example: $(17)_{10} = (10001)_2$ 5-digits number to base 2

$(17)_{10} = (11)_{16}$ 2-digits number to base 16

The total number of digits, $\lfloor \log_b n \rfloor + 1$

Addition and multiplication of numbers

Adding $2\ k$ -bit integers m and n .

What is the total number of bit operation required???

HW

Multiply $2\ k$ -bit integers m and n .

What is the total number of bit operation required???

HW

Primes

Prime: A positive integer $p > 1$ is called a prime iff it is divisible by 1 and the number itself.

Composite: A positive integer that is not prime is called a composite.

Theorem: The fundamental theorem of arithmetic

Every integer can be written as the product of primes (in order of nondecreasing size) in a unique way.

$$n = \pm p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

where p_i 's are k distinct primes and e_i 's are integers with $e_i > 0$.

Example: $96 = 2^5 \cdot 3^1$ $45 = 3^2 \cdot 5^1$

The Sieve of Eratosthenes

It is used to find all primes not exceeding a given positive integer n .

Example: List all primes upto 30

- keep 2, and then delete all integers that are divisible by 2.
- keep 3, and then delete all integers that are divisible by 3.
- keep 5, and then delete all integers that are divisible by 5.
- keep 7, and then delete all integers that are divisible by 7.
- keep 11, and then delete all integers that are divisible by 11.
- keep 13, and then delete all integers that are divisible by 13.
- continue this process

The leftover numbers are primes.

Composite

n be a composite number, then it has a prime divisor $\leq \sqrt{n}$.

Why is this true?

Composite

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Why is this true?

$n = x \cdot y$ then $x \leq \sqrt{n}$ or $y \leq \sqrt{n}$

A trivial algorithm:

Composite

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Why is this true?

$n = x \cdot y$ then $x \leq \sqrt{n}$ or $y \leq \sqrt{n}$

A trivial algorithm: Check whether a number is composite or not

- Check for any number $\leq \sqrt{n}$ that is a divisor of n or not.

Running time:

Composite

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Running time: $O(\sqrt{n})$

This is an exponential-time algorithm.

Why?

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$n = x \cdot y$ then $x \leq \sqrt{n}$ or $y \leq \sqrt{n}$

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Why?

Size of the input: $k = O(\log n)$, since the input is n

Running time: $O(2^k)$

Euclid's theorem

Theorem: Euclid's theorem There are infinitely many primes.

Proof:

- Assume that there are finite numbers of primes i.e. $p_1, p_2, \dots, p_n$
- Consider $t = p_1 \cdot p_2 \cdot \dots \cdot p_n + 1$
- Either t is a prime or t is a product of primes.
- If t is not prime, any p_i divides t .
- Then $p_i | (t - p_1 \cdot p_2 \cdot \dots \cdot p_n)$, implies $p_i | 1$
- So p_i does not divide t as it leaves 1 as remainder.
- Fundamental theorem of arithmetic says there exists a q that is a factor of t , as t is a product of primes.
- q must be different from p_i for $1 \leq i \leq n$

Marsene primes

The prime numbers of the form $2^p - 1$, where p is a prime, are called marsene primes.

Example: $3 = 2^2 - 1$ $31 = 2^5 - 1$

The largest known prime numbers are marsene primes

- The largest known prime number (07 Dec 2018)
 $2^{82589933} - 1$

It has 25,862,048 digits.

Distribution of primes

Theorem (The prime number theorem) The number of primes that does not exceed x can be approximated by $\frac{x}{\log x}$.

Twin primes

Twin primes are pairs of primes that differ by 2.

Example: 3 and 5 is twin prime.

17 and 19 is twin prime.

(29, 31), (101, 103)

Conjecture (Twin prime conjecture): There are infinitely many twin primes.

- The largest known twin prime till 2016 is

$$2,996,863,034,895 \times 2^{1,290,000} \pm 1$$

- It has 3,88,342 digits

Greatest common divisor (gcd)

gcd: Given a, b , where a and b are integers and $b > 0$, the greatest common divisor of a and b denoted by $\gcd(a, b)$ is the positive divisor of both a and b .

In other words, the largest integer d such that $d|a$ and $d|b$

Example: $\gcd(15, 27) = 3$ $\gcd(15, 45) = 15$

Relatively prime

Two integers a and b are relatively prime if $\gcd(a, b) = 1$.

Example: 15 and 17 are relatively prime as $\gcd(15, 17) = 1$

The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ for all $1 \leq i < j \leq n$

Finding gcd

We want to find the gcd of a and b , that is, $\gcd(a, b)$

- $a = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n} \quad e_i \geq 0$
- $b = p_1^{f_1} p_2^{f_2} \dots p_n^{f_n} \quad f_i \geq 0$
- $\gcd(a, b) = p_1^{\min\{e_1, f_1\}} p_2^{\min\{e_2, f_2\}} \dots p_n^{\min\{e_n, f_n\}}$

This method is not efficient.

Why???

For prime factorization, there is no efficient algorithm available.

Least common multiple (lcm)

lcm: The lcm of two positive integers a and b is the smallest positive integer that is divisible by both a and b and is denoted by $\text{lcm}(a, b)$.

In other words, $a|\ell$ and $b|\ell$ where ℓ is the smallest integer.

Finding lcm of a and b , that is, $\text{lcm}(a, b)$

- $a = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n} \quad e_i \geq 0$
- $b = p_1^{f_1} p_2^{f_2} \dots p_n^{f_n} \quad f_i \geq 0$
- $\text{lcm}(a, b) = p_1^{\max\{e_1, f_1\}} p_2^{\max\{e_2, f_2\}} \dots p_n^{\max\{e_n, f_n\}}$

Properties of gcd

- $\gcd(a, b) = \gcd(b, a)$
- $\gcd(a, a) = a$
- $\gcd(a, b) = \gcd(a - b, b)$
- $\gcd(0, a) = a$

Let a and b be two positive integers, then

$$ab = \gcd(a, b) \times \text{lcm}(a, b)$$

Euclid's algorithm

Euclid's algorithm for finding gcd

Input: Two non-negative integers a and b

Output: $\gcd(a, b)$

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Euclid(a, b)

while $b \neq 0$

$r \leftarrow a \bmod b$

$a \leftarrow b$

$b \leftarrow r$

return a

Euclid's algorithm

Euclid's algorithm for finding gcd

Input: Two non-negative integers a and b

Example: $\gcd(1296, 592) = 16$

Output: $\gcd(a, b)$

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```
Euclid( $a, b$ )  
while  $b \neq 0$   
     $r \leftarrow a \bmod b$   
     $a \leftarrow b$   
     $b \leftarrow r$   
return  $a$ 
```

Example: $\gcd(1296, 592) = 16$

$a = 1296$ $b = 592$
 $r \leftarrow 112$

Euclid's algorithm

Euclid's algorithm for finding gcd

Input: Two non-negative integers a and b

Output: $\gcd(a, b)$

Euclid(a, b)

while $b \neq 0$

$r \leftarrow a \bmod b$

$a \leftarrow b$

$b \leftarrow r$

return a

Example: $\gcd(1296, 592) = 16$

$a = 1296$ $b = 592$

$r \leftarrow 112$

$a = 592$ $b = 112$

$r \leftarrow 32$

$a = 112$ $b = 32$

$r \leftarrow 16$

$a = 32$ $b = 16$

$r \leftarrow 0$

$a = 16$ $b = 0$

Euclid's algorithm: correctness

Theorem: Let $a = bq + r$, where a , b , q , and r are integers with $r \geq 0$, then $\gcd(a, b) = \gcd(b, r)$.

Proof:

- Let $d = \gcd(a, b)$
- We prove that d is also a gcd of b and r
- $d = \gcd(a, b)$ implies $d|a$ and $d|b$
- $a = bq + r$ implies $a - bq = r$
- If $d|a$ and $d|b$ then $d|(a - bq)$, implies $d|r$
- $d|b$ and $d|r$
- d is a common divisor of b and r

gcd as a linear combination

Theorem (Bezout's theorem): If a and b are positive integers then there exists integers s and t such that $\gcd(a, b) = sa + tb$.

How to find s and t ???

Example: Let $a = 420$ and $b = 98$

$$a = 420 \quad b = 98$$

$$a = kb + r$$

$$420 = 4 \cdot 98 + 28$$

$$a = 98 \quad b = 28$$

$$98 = 3 \cdot 28 + 14$$

$$a = 28 \quad b = 14$$

$$28 = 2 \cdot 14 + 0$$

$$a = 14 \quad b = 0$$

$$\gcd(14, 0) = 14$$

We have to find two integers s and t such that,

$$\gcd(420, 98) = 14 = s \cdot 420 + t \cdot 98$$

gcd as a linear combination

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$$a = 28 \quad b = 14$$

$$28 = 2 \cdot 14 + 0$$

$$a = 14 \quad b = 0$$

$$\gcd(14, 0) = 14$$

We have to find two integers s and t such that,

$$\gcd(420, 98) = 14 = s \cdot 420 + t \cdot 98$$

$$\gcd(420, 98) = 14$$

$$14 = 98 - 3 \cdot 28$$

$$= 98 - 3 \cdot [420 - 4 \cdot 98]$$

$$= 98 - 3 \cdot 420 + 12 \cdot 98$$

$$= -3 \cdot 420 + 13 \cdot 98$$

Here, $s = -3$ and $t = 13$

Linear congruences

A congruence of the form

$$ax \equiv b \pmod{m}$$

where m is a positive integer and a and b are integers and x is an integer variable.

The solution of this congruence are all the integers x that satisfy it.

Multiplicative inverse

An integer $\bar{a}$ is called a multiplicative inverse of a modulo m if

$$a\bar{a} \equiv 1 \pmod{m}$$

Question: How multiplicative inverse is used to solve linear congruence??

$$ax \equiv b \pmod{m}$$

$$\Rightarrow \bar{a}ax \equiv \bar{a}b \pmod{m}$$

$$\Rightarrow x \equiv \bar{a}b \pmod{m}$$

Example

Find the multiplicative inverse of 5 modulo 21

Solution:

$$5 \cdot 1 \equiv 5 \text{ modulo } 21$$

$$5 \cdot 2 \equiv 10 \text{ modulo } 21$$

$$5 \cdot 3 \equiv 15 \text{ modulo } 21$$

$$5 \cdot 4 \equiv 20 \text{ modulo } 21$$

$$5 \cdot 5 \equiv 25 \text{ modulo } 21 \equiv 4 \text{ modulo } 21$$

$$5 \cdot 6 \equiv 30 \text{ modulo } 21 \equiv 9 \text{ modulo } 21$$

$$5 \cdot 7 \equiv 35 \text{ modulo } 21 \equiv 14 \text{ modulo } 21$$

$$5 \cdot 8 \equiv 40 \text{ modulo } 21 \equiv 19 \text{ modulo } 21$$

$\vdots$

$$5 \cdot 17 \equiv 85 \text{ modulo } 21 \equiv 1 \text{ modulo } 21$$

17 is the multiplicative inverse of 5 modulo 21

Multiplicative inverse: example

Find the multiplicative inverse of 3 modulo 6

Solution:

$$3 \cdot 1 = 3 \equiv 3 \text{ modulo } 6$$

$$3 \cdot 2 = 6 \equiv 0 \text{ modulo } 6$$

$$3 \cdot 3 = 9 \equiv 3 \text{ modulo } 6$$

$$3 \cdot 4 = 12 \equiv 0 \text{ modulo } 6$$

If a and m are relatively prime and $m > 1$, then a multiplicative modulo m exists. Furthermore, this inverse is unique.

Solving linear congruence

Find x such that $3x \equiv 7 \pmod{10}$

Solution: The inverse of $3 \pmod{10}$ is 7 as $3 \cdot 7 = 21 \equiv 1 \pmod{10}$

$$3x \equiv 7 \pmod{10}$$

$$x \equiv \bar{3}7 \pmod{10}$$

$$x \equiv 7 \cdot 7 \pmod{10}$$

$$x \equiv 49 \pmod{10}$$

$$x \equiv 9 \pmod{10}$$

Chinese remainder theorem

Chinese remainder theorem: Let $m_1, m_2, \dots, m_n$ be n pairwise relatively prime positive integers greater than 1 and $a_1, a_2, \dots, a_n$ be n arbitrary integers. Then the system

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\vdots \\x &\equiv a_n \pmod{m_n}\end{aligned}$$

has a unique solution modulo $m = m_1 \cdot m_2 \cdot \dots \cdot m_n$

Proof: Define $M_i = m_1 \times m_2 \times \dots \times m_{i-1} \times m_{i+1} \times \dots \times m_n$
 M_i is the product of all m_i 's excluding m_i .

- $\gcd(M_i, m_i) = 1$ H.W.
- By Bezout's identity, $\gcd(M_i, m_i) = s_i M_i + t_i m_i$
Implies $s_i M_i + t_i m_i = 1$ further implies $s_i M_i \equiv 1 \pmod{m_i}$

Chinese remainder theorem

- Let $x = a_1M_1s_1 + a_2M_2s_2 + \dots + a_nM_ns_n$ on mod m

We need to prove two things:

- x is a solution to the system.
 - Taking mod m_i on x we have,
$$x \equiv a_1M_1s_1 + a_2M_2s_2 + \dots + a_nM_ns_n \pmod{m_i}$$
$$x \equiv a_iM_is_i \pmod{m_i} \quad \text{every term has } m_i \text{ as a factor}$$
$$x \equiv a_i \pmod{m_i} \quad \text{as } s_iM_i \equiv 1 \pmod{m_i}$$
- x is a unique solution.
 - Assume that x and y are two solutions of the system.
 - $x = y = a_i \pmod{m_i}$
 - $x - y = 0 \pmod{m_i}$
 - $m_i | (x - y)$
 - $m | (x - y)$
 - $(x - y) \equiv 0 \pmod{m}$
 - $x = y$

Chinese remainder theorem: example

Example: Solve

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$x \equiv 4 \pmod{6}$$

Solution: $x \equiv a_1 M_1 s_1 + a_2 M_2 s_2 + a_3 M_3 s_3 \pmod{m}$

$$M_1 = 7 \times 6 = 42 \quad M_1 s_1 \equiv 1 \pmod{5} \quad M_1 = 42 \equiv 2 \pmod{5}$$

The multiplicative inverse to $M_1 \pmod{5}$ is 3 as $3 \cdot 2 \equiv 1 \pmod{5}$

Hence, $s_1 = 3$

$$\text{Similarly, } M_2 = 5 \times 6 = 30 \quad s_2 = 4 \quad M_2 s_2 \equiv 1 \pmod{7}$$

$$\text{Similarly, } M_3 = 5 \times 7 = 35 \quad s_3 = 5 \quad M_3 s_3 \equiv 1 \pmod{6}$$

$$x \equiv 3 \cdot 42 \cdot 3 + 2 \cdot 30 \cdot 4 + 4 \cdot 35 \cdot 5 \pmod{210}$$

Fermat's little theorem

Fermat's little theorem: If p is a prime then $a^{p-1} \equiv 1 \pmod{p}$ if $p \nmid a$.

That is $a^p \equiv a \pmod{p}$ for each integer a .

Example: Find $7^{222} \pmod{11}$

- $11 \nmid 7$
- From Fermat's little theorem, $7^{10} \equiv 1 \pmod{11}$
- $7^{222} = 7^{10 \cdot 22} \cdot 7^2 \pmod{11} \equiv 7^2 \pmod{11} \equiv 5 \pmod{11}$

Modular exponentiation

Computation: $b^n \bmod m$

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Computation: $b^n \bmod m$

A simple approach

- Successively computing $b^k \bmod m$ for $k = 1, 2, \dots, n$
 $b^{k+1} \bmod m = b(b^k \bmod m) \bmod m$

Modular exponentiation

Computation: $b^n \bmod m$

A simple approach

- Successively computing $b^k \bmod m$ for $k = 1, 2, \dots, n$
 $b^{k+1} \bmod m = b(b^k \bmod m) \bmod m$
- This approach is impractical as it requires $n - 1$ multiplications of integers and n might be huge.

Fast modular exponentiation

- $n = (a_{k1} \dots a_1 a_0)_2$

Fast modular exponentiation

- $n = (a_{k-1} \dots a_1 a_0)_2$
- $b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2^1 + a_0 \cdot 2^0} = b^{a_{k-1} \cdot 2^{k-1}} \cdot \dots \cdot b^{a_1 \cdot 2^1} \cdot b^{a_0 \cdot 2^0}$.

Fast modular exponentiation

- $n = (a_{k-1} \dots a_1 a_0)_2$
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- Compute $b, b^2, (b^2)^2 = b^4, (b^4)^2 = b^8, \dots, (b^{2^{k-1}})^2 = b^{2^k}$.
- Multiply the terms b^{2^j} in this list, where $a_j = 1$.
- Take $\text{mod } m$ after each multiplication if possible

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- Multiply the terms b^{2^j} in this list, where $a_j = 1$.
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Example: Compute 3^{11}

Fast modular exponentiation

- $n = (a_{k-1} \dots a_1 a_0)_2$
- $b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2^1 + a_0 \cdot 2^0} = b^{a_{k-1} \cdot 2^{k-1}} \cdot \dots \cdot b^{a_1 \cdot 2^1} \cdot b^{a_0 \cdot 2^0}$.
- Compute $b, b^2, (b^2)^2 = b^4, (b^4)^2 = b^8, \dots, (b^{2^{k-1}})^2 = b^{2^k}$.
- Multiply the terms b^{2^j} in this list, where $a_j = 1$.
- Take $\text{mod } m$ after each multiplication if possible

Example: Compute 3^{11}

- $11 = (1011)_2$

Fast modular exponentiation

- $n = (a_{k-1} \dots a_1 a_0)_2$
- $b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2^1 + a_0 \cdot 2^0} = b^{a_{k-1} \cdot 2^{k-1}} \cdot \dots \cdot b^{a_1 \cdot 2^1} \cdot b^{a_0 \cdot 2^0}$.
- Compute $b, b^2, (b^2)^2 = b^4, (b^4)^2 = b^8, \dots, (b^{2^{k-1}})^2 = b^{2^k}$.
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- Take $\text{mod } m$ after each multiplication if possible

Example: Compute 3^{11}

- $11 = (1011)_2$
- $3^{11} = 3^8 3^2 3^1$.
- $3^2 = 9, 3^4 = 9^2 = 81$, and $3^8 = (81)^2 = 6561$.

Fast modular exponentiation

- $n = (a_{k-1} \dots a_1 a_0)_2$
- $b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2^1 + a_0 \cdot 2^0} = b^{a_{k-1} \cdot 2^{k-1}} \cdot \dots \cdot b^{a_1 \cdot 2^1} \cdot b^{a_0 \cdot 2^0}$.
- Compute $b, b^2, (b^2)^2 = b^4, (b^4)^2 = b^8, \dots, (b^{2^{k-1}})^2 = b^{2^k}$.
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- $3^{11} = 3^8 3^2 3^1$.
- $3^2 = 9, 3^4 = 9^2 = 81$, and $3^8 = (81)^2 = 6561$.
- $3^{11} = 3^8 3^2 3^1 = 6561 \cdot 9 \cdot 3 = 177147$.

$b \text{ mod } m, b^2 \text{ mod } m, b^4 \text{ mod } m, \dots, b^{2^j} \text{ mod } m$ and multiplies together those terms $b^{2^j} \text{ mod } m$ where $a_j = 1$,

Fast modular exponentiation

mod_expo(b, n, m)

$n = (a_{k-1} \dots a_1 a_0)_2$

$x \leftarrow 1$

$pr \leftarrow b \bmod m$

for $i = 0$ to $k - 1$

 if $a_i = 1$ then

$x := (x \cdot pr) \bmod m$

$pr := (pr \cdot pr) \bmod m$

return x { x equals $b^n \bmod m$ }

Fast modular exponentiation

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Fast modular exponentiation

Compute 3^{11} $11 = (1011)_2$

$$x = 1$$

$$pr = 3 \bmod 5$$

mod_expo(b, n, m)

$$n = (a_{k-1} \dots a_1 a_0)_2$$

$$x \leftarrow 1$$

$$pr \leftarrow b \bmod m$$

for $i = 0$ to $k - 1$

 if $a_i = 1$ then

$$x := (x \cdot pr) \bmod m$$

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Compute 3^{11} $11 = (1011)_2$

$x = 1$

$pr = 3 \bmod 5$

$i = 0, a_0 = 1$

$x = 1 \cdot 3 = 3 \bmod 5$

$pr = 3 \cdot 3 = 9 \bmod 5 = 4$

Fast modular exponentiation

mod_expo(b, n, m)

$n = (a_{k-1} \dots a_1 a_0)_2$

$x \leftarrow 1$

$pr \leftarrow b \bmod m$

for $i = 0$ to $k - 1$

 if $a_i = 1$ then

$x := (x \cdot pr) \bmod m$

$pr := (pr \cdot pr) \bmod m$

return x { x equals $b^n \bmod m$ }

Compute 3^{11} $11 = (1011)_2$

$x = 1$

$pr = 3 \bmod 5$

$i = 0, a_0 = 1$

$x = 1 \cdot 3 = 3 \bmod 5$

$pr = 3 \cdot 3 = 9 \bmod 5 = 4$

$i = 1, a_1 = 1$

$x = 3 \cdot 4 = 12 \bmod 5 = 2$

$pr = 4 \cdot 4 = 16 \bmod 5 = 1$

Fast modular exponentiation

mod_expo(b, n, m)

$n = (a_{k-1} \dots a_1 a_0)_2$

$x \leftarrow 1$

$pr \leftarrow b \bmod m$

for $i = 0$ to $k - 1$

 if $a_i = 1$ then

$x := (x \cdot pr) \bmod m$

$pr := (pr \cdot pr) \bmod m$

return x { x equals $b^n \bmod m$ }

Compute 3^{11} $11 = (1011)_2$

$x = 1$

$pr = 3 \bmod 5$

$i = 0, a_0 = 1$

$x = 1 \cdot 3 = 3 \bmod 5$

$pr = 3 \cdot 3 = 9 \bmod 5 = 4$

$i = 1, a_1 = 1$

$x = 3 \cdot 4 = 12 \bmod 5 = 2$

$pr = 4 \cdot 4 = 16 \bmod 5 = 1$

$i = 2, a_2 = 0, \quad x = 2 \bmod 5$

$pr = 1 \cdot 1 = 1 \bmod 5$

Fast modular exponentiation

mod_expo(b, n, m)

$n = (a_{k-1} \dots a_1 a_0)_2$

$x \leftarrow 1$

$pr \leftarrow b \bmod m$

for $i = 0$ to $k - 1$

 if $a_i = 1$ then

$x := (x \cdot pr) \bmod m$

$pr := (pr \cdot pr) \bmod m$

return x { x equals $b^n \bmod m$ }

Compute 3^{11} $11 = (1011)_2$

$x = 1$

$pr = 3 \bmod 5$

$i = 0, a_0 = 1$

$x = 1 \cdot 3 = 3 \bmod 5$

$pr = 3 \cdot 3 = 9 \bmod 5 = 4$

$i = 1, a_1 = 1$

$x = 3 \cdot 4 = 12 \bmod 5 = 2$

$pr = 4 \cdot 4 = 16 \bmod 5 = 1$

$i = 2, a_2 = 0, \quad x = 2 \bmod 5$

$pr = 1 \cdot 1 = 1 \bmod 5$

$i = 3, a_3 = 1$

$x = 2 \cdot 1 = 2 \bmod 5$

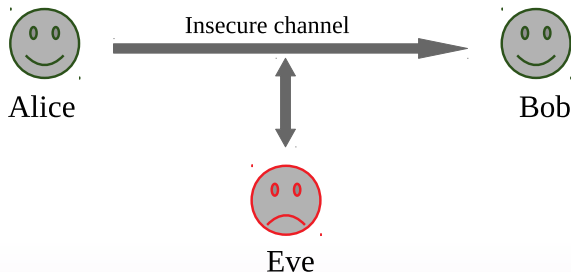
$pr = 1 \cdot 1 = 1 \bmod 5$

Cryptography

Cryptography: Cryptography is the art (and sometimes science) of secret writing.

- Cryptographers create ciphers - - - **Cryptography**
- Cryptanalyst break ciphers - - - **Cryptanalysis**
- **Cryptology = cryptography + cryptanalysis**

Cryptography goals



- Alice wants to send message to Bob through an insecure channel.
- Eve tries to capture the message and modify it.
- Alice prevents Eve from understanding the message. (Encryption)
- Bob must understand that the unchanged message comes from Alice, not from Eve. (Authentication)

A few terminology

Cryptography: It is a process of encoding and decoding messages.

Cipher: A method of encoding message

Plaintext: The message that needs to be encoded

Ciphertext: The encoded message

Encryption: The process of encoding messages

Decryption: The process of decoding messages

Cryptosystem

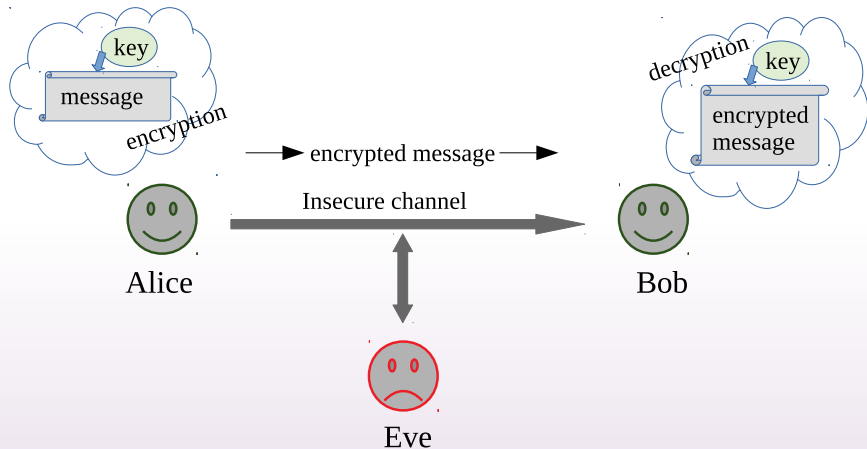
Cryptosystem: A cryptosystem is a 5-tuple (E, D, M, K, C) .

- E is an encryption algorithm
- D is a decryption algorithm
- M is the set of plaintexts
- K is the set of keys
- C is the set of ciphertexts

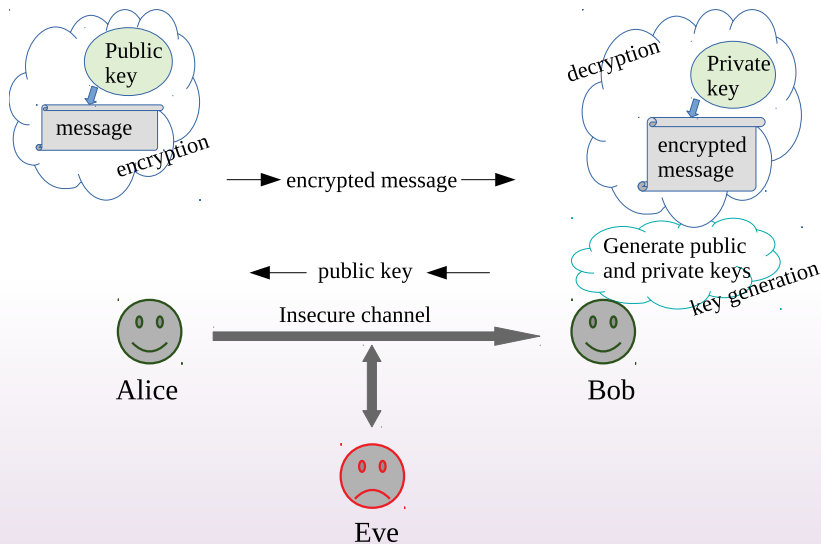
Encryption: $E : M \times K \rightarrow C$

Decryption: $D : C \times K \rightarrow M$

Symmetric (secret) key



Asymmetric (public) key



Bob only knows the private key.

Caesar cipher

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Encoding: Let x is encoded to c using the following rule:

$$c = (x + 17) \bmod 26$$

HELLO - - - - - > YVCCF

H $\rightarrow$ 7 $\rightarrow$ 24 $\bmod 26 \rightarrow$ 24 $\rightarrow$ Y

E $\rightarrow$ 4 $\rightarrow$ 21 $\bmod 26 \rightarrow$ 21 $\rightarrow$ V

L $\rightarrow$ 11 $\rightarrow$ 28 $\bmod 26 \rightarrow$ 2 $\rightarrow$ C

L $\rightarrow$ 11 $\rightarrow$ 28 $\bmod 26 \rightarrow$ 2 $\rightarrow$ C

O $\rightarrow$ 14 $\rightarrow$ 31 $\bmod 26 \rightarrow$ 5 $\rightarrow$ F

Decoding:

$$x = (c - 17) \bmod 26$$

F $\rightarrow$ 5 $\rightarrow$ -12 $\bmod 26 \rightarrow$ (26 - 12) $\bmod 26 \rightarrow$ 14 $\rightarrow$ O

RSA cryptosystem

- The RSA cryptosystem was developed in 1978 by Ron Rivest, Adi Shamir, and Leonard Adleman.
- RSA is an example of a public-key cryptosystem, and these are widely used today.
- It has two phases - - - Key generation and encryption and decryption

RSA: key generation

- Choose 2 (large) primes, e.g., $p = 17$, $q = 19$.
Normally, these would be several hundred digits in length.
- Let $n = pq = 17 \cdot 19 = 323$.
Factoring such a large n is basically impossible. Only **you** know p and q !
- Let $A = \varphi(n) = (p - 1)(q - 1) = 16 \cdot 18 = 288$.
Without knowing how to factor n , computing $\varphi(n)$ is impossible.
- Pick $E < \varphi(n)$ such that $\gcd(E, \varphi(n)) = 1$. [Let's pick $E = 95$].
Your **public key** is $(n, E) = (323, 95)$
- Compute your **private key**, $D = E^{-1} \bmod \varphi(n)$, i.e., the solution to $Ex \equiv 1 \bmod \varphi(n)$.

RSA: encryption and decryption

Encryption: A message M needs to be sent.

The (public) encryption function is

$$c = e(M) = M^E \bmod n, [e(M) = M^{95} \bmod 323].$$

Send c to Bob

Decryption: The decryption function, known only to you, is (modulo n)

$$M = d(c) \equiv c^D \bmod n, [d(c) = c^{191} \bmod 323]$$

You choose $p = 17$, $q = 19$,

public key $(n, E) = (323, 95)$.

Compute the private key $D = E^{-1} = 191$. (We'll learn how to do this.

Let $M = 86$

Encryption: $e(86) = 86^{95} \equiv 307 \bmod 323$

307 sent to Bob

Decryption: $d(307) = 307^{191} \bmod 323 \equiv 86 \bmod 323$

RSA: Three major computation

We need to learn how to do the following

- 1 Find $E \in \mathbb{N}$ such that $\gcd(E, \varphi(n)) = 1$. [e.g., $\gcd(E, 288) = 1$.]
Most systems use $E = 65537$.
- 2 Solve $Ex \equiv 1 \pmod{\varphi(n)}$. [e.g., solve $95x \equiv 1 \pmod{288}$.]
Extended Euclidean algorithm.
- 3 Compute M^E and c^D modulo n . [e.g., $86^{95} \pmod{n}$ and $307^{191} \pmod{n}$.]
“Fast modular exponentiation”, uses method of repeated squaring

Computing E s.t. $\gcd(E, \varphi(n)) = 1$

- In the example $n = pq = 17 \cdot 19 = 323$, $\varphi(n) = 16 \cdot 18 = 288$, and as the message recipient, one needed to find E such that $\gcd(E, 288) = 1$.
- For small n , this is easy: factor 288 and pick a number with no common prime factors.
- In practice, $\varphi(n)$ is too large to factor. But any prime that does not divide $\varphi(n) = (p-1)(q-1)$ will work.
- Guessing and checking will yield a prime rather quickly.
- A particularly nice choice of E would be:
 - prime [makes it easier to verify that $\gcd(E, \varphi(n)) = 1$],
 - of the form $2^n + 1$, because this is 1000...001 in binary.
- The only primes of the form $2^n + 1$ also have the form $2^{2^k} + 1$, called Fermat primes.
- The only known Fermat primes are 3, 5, 17, 257, 65537. As such, in practice, $E = 2^{2^4} + 1 = 65537$ is usually used for encryption.
- In the very slim chance that 65537 divides $\varphi(n) = (p-1)(q-1)$, then go back and pick a new p and q .

How to solve $Ex \equiv 1 \pmod{\varphi(n)}$

- We can solve the $Ex \equiv 1 \pmod{\varphi(n)}$ using the extended Euclidean algorithm.

Let's solve $95x \equiv 1 \pmod{288}$.

$$\gcd(288, 95) = 1$$

$$288 = 3 \cdot 95 + 3$$

$$95 = 31 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$1 = 1 \cdot 1 + 0$$

$$1 = 3 - 1 \cdot 2$$

$$= 3 - 1 \cdot [95 - 3 \cdot 31]$$

$$= -1 \cdot 95 + 3 \cdot 32$$

$$= -1 \cdot 95 + [288 - 95 \cdot 3] \cdot 32$$

$$= 32 \cdot 288 + (-97) \cdot 95$$

We conclude that: $\gcd(288, 95) = 1 = 288(32) + 95(97)$.

From this, we can solve

$$95x \equiv 1 \pmod{288} \Rightarrow x = -97 \equiv 191 \pmod{288}.$$

Computing M^E and c^D modulo $n = pq$.

Use fast modular exponentiation compute $86^{95} \bmod 323$.

$$95 = 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = (1011111)_2.$$

$$x = 1 \quad pr = 86 \bmod 323$$

$$i = 0, a_0 = 1$$

$$x = 1 \cdot 86 = 86 \bmod 323$$

$$pr = 86 \cdot 86 = 290 \bmod 323$$

$$i = 1, a_1 = 1$$

$$x = 86 \cdot 290 = 24940 \bmod 323 = 69$$

$$pr = 290 \cdot 290 = xxxx$$

$$\bmod 323 = 120$$

$$i = 2, a_2 = 1,$$

$$x = 69 \cdot 120 = xxxx \bmod 323 = 205$$

$$pr = 120 \cdot 120 = xxxx$$

$$\bmod 323 = 188$$

$$i = 3, a_3 = 1$$

$$x = 205 \cdot 188 = xxxx \bmod 323 = 103$$

$$pr = 188 \cdot 188 = xxxx$$

$$\bmod 323 = 137$$

$$i = 4, a_4 = 1,$$

$$x = 103 \cdot 137 = xxxx \bmod 323 = 222$$

$$pr = 137 \cdot 137 = xxxx \bmod 323 = 35$$

$$i = 5, a_5 = 0,$$

$$x = 222$$

$$pr = 35 \cdot 35 = xxxx \bmod 323 = 256$$

$$i = 6, a_6 = 1,$$

$$x = 222 \cdot 256 = xxxx \bmod 323 = 307$$

This is called the method of repeated squaring. Clearly, things are (slightly) easier using $E = 65537 = (1000 \dots 0001)_2$.

Correctness of RSA

- We need to show that $c^D \bmod n$ is equal to M
- $c^D \equiv M^{ED} \bmod n \equiv M^{ED} \bmod n$
- We have $ED \equiv 1 \bmod \varphi(n)$, $ED = t \cdot \varphi(n) + 1$ for some integer t
- $c^D = M^{ED} \equiv M^{t\varphi(n)+1} \bmod n \equiv M^{t(p-1)(q-1)+1} \bmod pq$
- If we prove
 - 1 $c^D \equiv M \bmod p$
 - 2 $c^D \equiv M \bmod q$

Apply Chinese remainder theorem on (1) and (2), we get

$$c^D \equiv M \bmod pq$$

The solution is unique

- Fermat's little theorem: For prime p and $\gcd(a, p) = 1$ then $a^{p-1} \equiv 1 \bmod p$.
- By Fermat's little theorem we get
 - 1 $c^D \equiv M \cdot M^{(p-1)^{t(q-1)}} \bmod p \equiv M \cdot 1^{t(q-1)} \bmod p$
 - 2 $c^D \equiv M \cdot M^{(q-1)^{t(p-1)}} \bmod q \equiv M \cdot 1^{t(p-1)} \bmod q$